



NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech (MMD) (CBCS) Degree Examinations

December 2024 – January 2025

22MMD102 – FINITE ELEMENT METHODS AND ITS APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

1. a) Define FEM. With the block diagram explain general steps in FEM.

Marks	BT*	CO*	PO*
3	L*2	1	1
- b) Derive the stiffness matrix for 1D bar element by Direct Method.

3	L2	1	1
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- c) What are "Banded matrices". With example explain Half band width method of storing the matrix in computer memory.

4	L2	1	1
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- d) What is minimum potential energy? and derive the minimum potential energy expression for a 3D elastic subjected to Body force, Traction force, and Point load.

10	L2	1	1
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2. a) Explain the discretization process. Sketch the different types of elements 1D, and 2D, 3D elements used in the finite element analysis.

10	L2	1	1
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- b) Define Principle of minimum potential energy. Find $\{F\} = [K]\{Q\}$ relationship for the Spring system shown in Fig.2b by applying minimum potential energy principle.

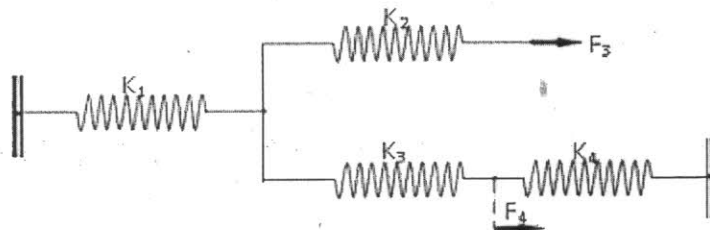


Fig 2b

10	L3	1	2
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Unit – II

3. a) Derive the expression of shape functions for a horizontal 1D quadratic element.

12	L3	2	1
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- b) Write a note on Lagrange Elements and Serendipity Elements with an example.

8	L2	2	1
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4. a) Explain the simplex, complex, and multiplex elements.

07	L2	2	1
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- b) For the two bar truss element shown in Fig 4b, determine the nodal displacements, stress in each elements and the reaction at support. Take $A = 200 \text{ mm}^2$, $E = 200 \text{ GPa}$.

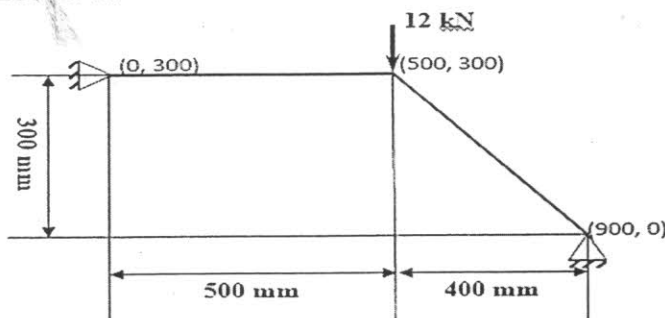


Fig 4b

13	L3	2	2
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Unit – III

5. a) Derive strain-displacement matrix for 2D triangular element.

12	L3	3	1
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P.T.O

- b) Determine the N_1 , N_2 and N_3 for a triangular element shown in Fig Q 5(b). The x-coordinate of the interior point P is 3.85 and y co-ordinate of the interior point p is 4.8

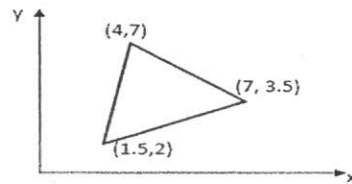


Fig 5b

08 L3 3 2

6. a) Derive strain-displacement matrix for an axisymmetric triangular element.
b) Write a note on 'axisymmetric problems'.

10 L3 3 1
10 L2 3 1

Unit – IV

7. a) For the beam shown in Fig.Q7 (a). Determine deflection at mid-span. Assume Young's Modulus for beam material to be 200GPa and the area moment of inertia of the cross-section of the beam as $4 \times 10^6 \text{ mm}^4$

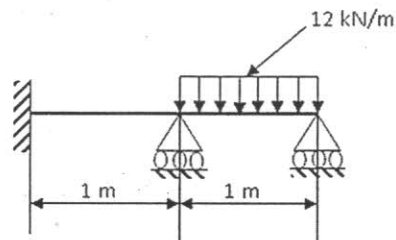


Fig 7a

10 L3 4 2

- b) Derive the hermit shape function of the beam element and draw the shape function variation over the beam element.
8. a) Derive the elemental stiffness matrix for the beam element.
b) A uniform cross-sectional beam is fixed at one end supported by a roller at the other end. A concentrated 20kN is applied at the mid length of the beam as shown in figure Fig Q8(b). Determine the deflection under load. Take $E = 210\text{GPa}$ and $I = 2500\text{mm}^4$

10 L3 4 1

08 L3 4 1

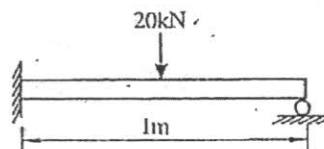


Fig 8b

12 L3 4 2

Unit – V

9. a) Determine the temperature distribution in a 1- Dimensional fin as shown in Fig Q 9(a). There is a uniform generation of heat inside the wall of 500W/m^3 . Diameter of the circular rod 0.02m.

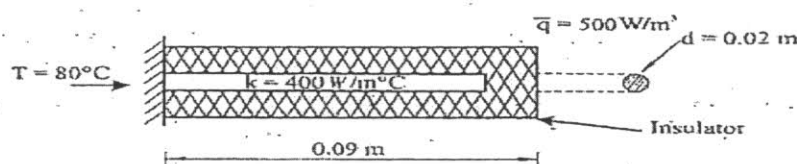


Fig Q9a

10 L3 5 2

- b) Briefly describe the boundary conditions used in heat transfer analysis.
10. a) Explain the characteristics of an algorithm.
b) Explain the steps involved in designing an Algorithm.

10 L2 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
